SWEEP rule on the KD-Toric code

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1 The Turns Idea.

The main problem we have is that, for now, the history is defined only for configurations that, at time t, see a non-trivial syndrome. What we can do is imagine that at each turn we flip the bits according to a fixed order. We think of the basic time unit as a turn, and each real-time iteration contains tn^d turns. Now, if a bulk of bits A is flipped at time t, then there must be a turn $\tau \leq tn^d$ such that at that turn there was a configuration that has a history.

Now we would like to describe what the history of that configuration looks like, and we do that by connecting the turn τ to the closest $t'n^d$. The idea is that we cann't ensure the shrinking lemma. Yet we can tell for sure that if there is expansion then its limited.

I have also the felling that the shrinking works, and the idea goes as follow, assume that $v \in \Gamma^{(t-1)}$ gets the maximum of L_i either we stop before flipping a bit that is 'supported' on v or not, in the first case, we get that any u that is added by v' might has $L_i(u) = L_i(v') < L_i(v)$, in the second case we have already handled.

The turn machinery makes sense only in the finite case. Yet, we can back to the Tree Separator Section, and show that with probability

$$n^d \cdot tn^d q^n$$

We don't have a big trees which implies independence. Then we can show that with probability:

$$n^d t (n^d)^2 q'^{\delta n}$$

There was no turn in which it had a configuration that was connected at the time, saw a syndrome, and had a side-length that behaved like $\Theta(n)$.

[COMMENT] The model has to redefined again such it will be clear that outcomes are possibles running of the system. Currently this can not be inferred.

Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, in different space-time locations. Each space-time location, associated with a gate, encodes the time of the application and the qubits in the support of the gate. Formally

Definition 1.1 (Quantum Circuit)**.** A space-time location l is a tuple that encodes in its first coordinate time, and in its second coordinate a subset of integers referring to a support of (q)bits. For a collection of quantum gates $\mathcal{G} = \{g \colon L(\mathcal{H}_2^{\otimes \Delta}) \to L(\mathcal{H}_2^{\otimes \Delta})\}$, and integers $w, d \in \mathbb{N}$ a quantum circuit C, with depth d and width w, is a collection of tuples $\{u_i = (g_i, l_i)\}$, where u_i are the gates of C and each u_i is composed of a gate $g_i \in \mathcal{G}$ and a space-time location $l_i \in \mathbb{Z}_d \times \mathbb{Z}_w^{\Delta}$. The gates of C have to satisfy the constraint that for every two gates with the same time coordinate, there cannot be a non-empty intersection in their space coordinate, namely for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies $l_{i,2} \cap l_{j,2} = \emptyset$. We define the t-cut of C to be $C|_t := \{u_i = (g_i, l_i) \colon u_i \in C \text{ and } l_{i,1} = t\}$.

To define the running of a quantum circuit C on a given (mixed) quantum state, we will need to have the notion of the acyclic directed graph associated with C.

Definition 1.2 (Associated Computation DAG)**.** Let $C = \{u_i\}$ be a quantum circuit with depth d and width w. We define the associated computation DAG of C, denoted by π_C, to be the acyclic directed graph $\pi_C = (\mathcal{U}, E)$ as follows:

1. The vertex set $\mathcal{U}$ is the gate set of C, namely $\mathcal{U} = \{u_i\}$.

2. For $u_i = (g_i, l_i), u_j = (g_j, l_j)$, There is an edge $\{u_i, u_j\} \in E$ if and only if:

$$j \in \arg\min_j \{l_{j,1} : l_{j,1} > l_{i,1} \text{ and } l_{j,2} \cap l_{i,2} \neq \emptyset\}$$

   Note that there might be multiple indices that get the minimum, so the out degree might be greater than one.

In addition, suppose that $J = \{j_1, j_2, \ldots, j_k\} \subset [|C|]$ is an order of the indices of the gates in C according to the topological order induced by π_C. Then, we say that J is an order of indices that agrees with π_C and define the series of circuit $C_J = \{C_{J,i}\}$ as follows:

1. The first element: $C_{J,1} = \{u_{j_1}\}$

2. Then, $C_{J,i+1} = C_{J,i} \cup \{u_{j_i}\}$

Definition 1.3. Let ρ be a density matrix over w qubits, and let $C = \{u_i\}$ be a quantum circuit with depth d and width w. For any density matrix σ, quantum gate $g \in \mathcal{G}$, and location l, the application of the gate-location tuple (g, l) on σ, denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ, denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}$ on ρ, according to a topological order induced by π_C.

We say that $\rho_1, \rho_2, \ldots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u'_i \circ \rho_i$, where $\{u'_i\}$ is a prefix of a topological ordering of $\{u_i\}$. If $\{u'_i\}$ contains all the gates in $\{u_i\}$, then we call $\rho_1, \rho_2, \ldots, \rho_k$ a running.

Definition 1.4 (Associated Computation DAG)**.** Let $C = \{u_i\}$ be a quantum circuit with depth d and width w. For a set of fault locations $\mathcal{E} \subset [d] \times [w]$, we define the quantum circuit $C[\mathcal{E}] = \{u'_i\}$ to be the result of the following map:

$$u_i = (g_i, (l_{i,1}, l_{i,2})) \mapsto \begin{cases} (g_i, (2l_{i,1}, l_{i,2})) & u_i \in C \\ (g_i, (2l_{i,1}-1, l_{i,2})) & u_i \in \mathcal{E} \end{cases}$$

We will denote the associated computation DAG of C affected by $\mathcal{E}$ as $\pi[C, \mathcal{E}]$.

Definition 1.5 (Faults Propagation to a Step τ)**.** For a quantum circuit $C = \{u_i\}$, with depth d, an order $J = \{j_1, j_2, \ldots, j_k\} \subset [|C|]$ that agrees with π_C, and gate u_i in C, a propagation of a gate $u_i \in C$ to step τ is the quantum operator $X\colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$ obtained by the follow process:

1. First, check if i precedes τ in J. If not, then return $X = I$. Otherwise, denote $j_l = i$, and look for an operator X_1 such that $u_{j_l} u_{j_{l+1}} = u_{j_{l+1}} X_1$.
2. Then, for $k \leq \tau - l$, look for an operator $X_k\colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$, such that $X_{k-1} u_{j_{l+1}} = u_{j_{l+1}} X_k$.
3. At the final stage of the process, namely where $k = \tau - l$, we define $X := X_k$ to be the propagation of u_j to step τ.

For a quantum circuit C, a set of faults $\mathcal{E} = \{e_i\}$, an order $J \subset [|C[\mathcal{E}]|]$ that agrees with $\pi_{C[\mathcal{E}]}$, and a step $\tau \in J$, denote by $X^\tau_{e_i}$ the propagation of $e_i \in \mathcal{E}$ in $C[\mathcal{E}]$ to step τ. Let $j'_1, j'_2, \ldots, j'_l$ be the indices of the faults in $\mathcal{E}$ such that their corresponding positions in J precede τ, ordered according to the reverse topological order induced by $\pi_{\mathcal{E}}$. The propagated error to step τ is defined by

$$X^\tau := \prod_i X^\tau_{e_{j'_i}}$$

Definition 1.6 (Deviation)**.** For a quantum circuit C, a set of faults $\mathcal{E}$, and step τ, let X^τ be the propagated error at step τ. We define the deviation of $C[\mathcal{E}]$ from C at step τ by the set of indices on which X^τ acts non-trivially.

Definition 1.7. Using the same notations as in Definition 1.6, for a set of unitary operators $\mathcal{S} = \{s\colon \mathcal{H}_2^w \to \mathcal{H}_2^w\}$, we define the propagated error up to stabilizers in $\mathcal{S}$ as $X^\tau[S] := \min_{s \in \mathcal{S}}\{w(sX)\}$. Then, we define the deviation of $C[\mathcal{E}]$ from C at step τ up to stabilizers in $\mathcal{S}$ to be the subset of coordinates on which $X^\tau[S]$ acts non-trivially.

Definition 1.8 (Generalized Tanner Graph)**.** For registers a and b, their generalized Tanner graph is their 'union' $\mathcal{T}_a \cup \mathcal{T}_b$. This allows us to define the Tanner graph of a quantum circuit C, at step τ, by stacking together Tanner graphs. This also allows us to define the Tanner graph if a register. rewrite. add definition for a register R of quantum circuit C.

Definition 1.9 (Clustered Deviation)**.** For a quantum circuit C, a register R of C, and a set of faults $\mathcal{E}$, let $\mathcal{T}_R$ be the Tanner graph of R, and let $\mathcal{D}_\tau$ be the deviation of $C[\mathcal{E}]$ from C on R at step τ. Denote by $\mathcal{T}_R|_{\mathcal{D}_\tau}$ the subgraph of $\mathcal{T}_R$ induced by the restriction of the left vertices (bits) to the indices in $\mathcal{D}_\tau$. Let us abuse and extend the connection notion to Tanner graphs by defining that two left vertices are connected if and only if they share a right neighbor. We define the deviation clusters at step τ to be the connected components of $\mathcal{T}_R|_{\mathcal{D}_\tau}$. The size of each deviation cluster is the number of right vertices it contains.

Claim 1.10. Let C be a quantum circuit composed of gates with bounded fan degree $\Delta \in \mathbb{N}$. In addition, let R be a quantum register encoded by a quantum error correction code $\mathcal{C}$ with Tanner graph $\mathcal{T}$ that has left and right degrees Δ_1 and Δ_2. For any τ, denote by $Y_1^{(\tau)}, Y_2^{(\tau)}, \ldots, Y_l^{(\tau)}$ the deviation clusters at step τ. Assume that at step τ_1 there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}| > \xi$. Then there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z, of size at least:

$$|Z| \geq \frac{1}{\Delta(1 + \Delta_1\Delta_2)}|Y_i^{(\tau_1)}|$$

Proof. □

Corollary 1.11. Assume that at step τ_1 there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}| > \xi$. Then there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z, of size at least:

Claim 1.12. Let $c \in \mathbb{R}$ be a constant, and let f be a function such,

$$|f\left(u_1, .., u_{k+1}\right) - f\left(u_1, .., u_k\right)| \leq c$$

Suppose that $f(u_1) = 0$,

Definition 1.13 (Unfolded configuration)**.** For a quantum circuit C, and a set of faults $\mathcal{E}$, Let $\pi[C, \mathcal{E}] = (\mathcal{U}, E)$ be their associated computation DAG. Let f For a running configuration $\omega \in \Omega$ we define the unfolded-configuration.